

DAILY PROGRESS REPORT

Tuesday, June 23, 2026

Internship — Sprint 1

Backend & API Setup
Shanza yasmeeen

DAILY SUMMARY

3

Tasks Assigned

3

Completed

1

Blockers

100%

Progress

TASK 1 — PROJECT SCAFFOLDING & ENVIRONMENT SETUP

1

GitHub Repository & Structure

Repo created with /backend (FastAPI), /frontend (React), /rag (chunking + embeddings), /docs (API contract, README).

✓DONE

2

Python Virtual Environment

fastapi, uvicorn, asyncpg, python-multipart, pypdf2 installed & verified

✓DONE

3

FastAPI Skeleton + SQLite DB

GET /health → {status: ok}; tables: files, subjects, units

✓DONE

Deliverables

- Empty main branch pushed to GitHub with clean folder structure
- uvicorn main:app confirmed running
- database.db created via sqlite3; schema shared in group chat

⚠️Blocker Encountered

Encountered an issue pushing to GitHub. Mentor intervened and resolved the configuration — push completed successfully. No code changes were required.

TASK 2 — API CONTRACT DRAFTING & REVIEW

1

Draft Full API Contract

Covered all 4 endpoints: upload, subjects, notes, chat API contract documented:

- Method
- URL
- Request body fields and types
- Response body fields and types
- Status codes

✓DONE

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2

Group Review by 13:00

Posted by 12:00; Jumana (AI & RAG Engineer) & Ayisha Saniyya (frontend engineer) reviewed on time

✓DONE

3

Commit api_contract.md

Committed before standup; no revisions needed

✓DONE

Endpoints Documented

- POST /upload-syllabus — multipart PDF upload, returns file_id
- GET /subjects/{file_id} — returns extracted subject list
- POST /generate-notes — triggers RAG note generation pipeline
- POST /chat — conversational Q&A over uploaded content

✓Review Outcome

No changes were flagged during peer review. The contract was accepted by both reviewers. Final api_contract.md committed to /docs before standup.

TASK 3 — DATABASE SCHEMA FINALISATION

1

Finalise schema.sql

Based on Jumana's ChromaDB design discussion

✓DONE

2

Commit to /docs

schema.sql pushed successfully after GitHub fix

✓DONE

Schema Details

- Incorporated ChromaDB vector collection references alongside relational tables, based on Jumana's (AI & RAG Engineer) design input
- Schema Details
-
- — Tables:
 - files — stores uploaded syllabus file metadata
 - subjects — stores subjects extracted from uploaded syllabi
 - units — stores units associated with each subject
-
- — ChromaDB Reference:
 - Collection: notes_chunks
 - Metadata fields: user_id, file_id, subject, chunk_id, source_file, unit, page_number
- schema.sql committed to /docs — aligned with API contract field names

END-OF-DAY NOTES

What Went Well

- All 3 tasks completed within the sprint day
- API contract accepted without revision — solid first-draft quality
- Mentor support resolved the GitHub blocker quickly

TESTING OBSERVATIONS

FastAPI Application — Health Check Verification

- The FastAPI application and health-check endpoint were functioning correctly.
- The endpoint was accessible and responding as expected at:
`http://127.0.0.1:8000/health`
- Since the initial application skeleton did not contain a root (/) endpoint, accessing the base URL:
`http://127.0.0.1:8000`
did not return a response page.
- This behavior was expected and confirmed that route configuration was functioning correctly — only explicitly defined routes are served.

CONCLUSION

All assigned Sprint 1 backend tasks were completed successfully. The project structure, FastAPI environment, database schema, API contract, and documentation were finalized and committed to GitHub. The backend foundation is now ready for the next phase of development.